Lateral analysis - Level 2 - Undamped natural frequency analysis

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1 Purpose.

This document is related to the dynamic analysis requirements for rotordynamic pump, especially to lateral analysis.

The procedure to define what level of detail for the dynamic analysis is required to be in accordance with good practice of pump design is described in G.10.00.09 and *ANSI/HI 9.6.8-2014*: if calculated RUN is higher than 20 and less equal to 50, a **Level 2 – Intermediate method - Lateral rotordynamic analysis - Undamped natural frequency analysis** is required.

This document describes how to perform an undamped natural frequency (lateral) analysis.

2 Abbreviations.

BB between bearings (API610)

DOF degree of freedom

FEA finite element analysis

OH overhung (API610)

R risk number (ANSI HI 9.6.8-2014)

RUN R multiplied by U (ANSI HI 9.6.8-2014)

U uncertainty value (ANSI HI 9.6.8-2014)

ROT_x rotational degree of freedom around x axis in cartesian coordinate system

ROT_y rotational degree of freedom around y axis in cartesian coordinate system

ROT_z rotational degree of freedom around z axis in cartesian coordinate system

U_x translational degree of freedom in x direction in cartesian coordinate system

U_y translational degree of freedom in y direction in cartesian coordinate system

U_z translational degree of freedom in z direction in cartesian coordinate system

3 Scope.

This procedure is applicable to OH and BB pumps with antifriction bearings (one or two stage).

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4 General procedure: description.

A lateral rotordynamic analysis is performed to assess the potential for high vibration, associated degradation, and eventually equipment failure after installation. Excessive lateral vibration can lead, or contribute, to the following types of problems:

- accelerated wear at internal wear ring clearances and loss of performances;
- rubs at internal clearance locations;
- accelerated bearing wear;
- excitation of high vibration associated with system structural modes, leading to fatigue of structural components.

Lateral critical speed analysis is performed for the combined pump rotating components (e.g., shaft, impellers, sleeves, coupling half, etc…) up to the first flexible shaft coupling (including half coupling), or for the entire pump-driver rotor system if a rigid coupling is used.

The entire pump rotor should be included in the model:

- shaft cross-sectional inertial and mass properties are calculated from the length and outer/inner diameter of each shaft section;
- impellers are modelled as added mass and rotational/transverse inertia referenced at shaft location;
- gyroscopic effects are included by the action of the component polar and transverse mass moments of inertia[1];
- a flexible coupling may isolate the lateral vibration between pump and driver[2];
- mathematical representation of shaft element should include not only bending stiffness, but also transverse shear stiffness.

For OH and BB pumps with antifriction bearings (one or two stages) an undamped natural frequency analysis is allowed for acceptance, although damped is most relevant for most pumping applications since seals has strong effect both on system damping and critical speeds.

For BB pumps damped natural frequency analysis is preferred: this increases the accuracy of the analysis by accounting for the Lomakin effect.

[1] This is relevant for OH pumps because the impeller produces a restoring moment when whirling in a deflected position.

[2] The pump rotor hub plus half spacer (if present) weight/inertia must be lumped at its center of gravity.

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5 Methodology, analysis and interpretation of results.

Methodology.

Software/Tools:

- general purpose FEA programs or specialty rotordynamic programs;
- software should account for gyroscopic effects.

The tool described in this procedure is:

- DyRoBes

Model.

Mass-elastic model of pump rotor only (low lateral stiffness of flexible disc type couplings decouple pump rotor from motor rotor with regards to lateral rotordynamic behavior).

Shaft segments modelled with beam and pipe elements.

Concentrated mass modelled as with lumped mass elements.

Elements have four DOF:

- translations in both lateral directions (horizontal U_x and vertical U_y; axial U_z is constrained at the trust bearing) and rotations around lateral axes (ROT_x and ROT_y; torsional ROT_z is constrained at all nodes);
- impellers and pump coupling half (pump coupling hub and half coupling spacer) are generally considered as lumped mass effects (mass and inertia) without stiffness contribution
    - impellers do not significantly contribute to the overall rotor stiffness in cases of a loose fit on the shaft;
    - impeller mounted with an interference fit may contribute to the overall rotor stiffness, depending on the level of interference (bracketing calculations may be performed to evaluate the influence of the interference fit on the rotor rigidity);
- shaft sleeves, rotating portion of the mechanical seal, etc., are considered as distributed mass effects (no stiffness contribution);
- interaction forces from annular seals and impeller/casing interaction are not considered (only liquid effects are from fluids contained within impeller passageways, and fluids displaced by wetted component volume).

Radial bearings are the only rotor support. Bearing stiffness and bearing support stiffness are both infinite (rigid bearings).

Analysis.

Undamped natural frequency analysis (modal).

Rotor speeds: minimum, rated and maximum continuous operating speed (one speed in case of fixed speed application).

Interpretation of results.

A Campbell diagram is typically not strictly necessary for this type of analysis.

Results of the natural frequency analysis (solution of the eigenvalue problem) are sufficient: this includes mode shape (eigenvector) and associated natural frequency (eigenvalue).

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6 Acceptance Criteria.

In case the first dry natural frequency (first dry “critical speed”) is at least 20% above the maximum continuous operating speed (30% in case rotor is designed to run dry), then the rotor is considered to be classically rigid (ref. API 610).

Classically rigid horizontal rotors are expected to have design integrity with regards to lateral rotordynamic behavior.

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7 Modelling recommendations.

7.1 Shaft modelling.

One of the most common questions about shaft modeling is how many elements or stations are required to get good results.

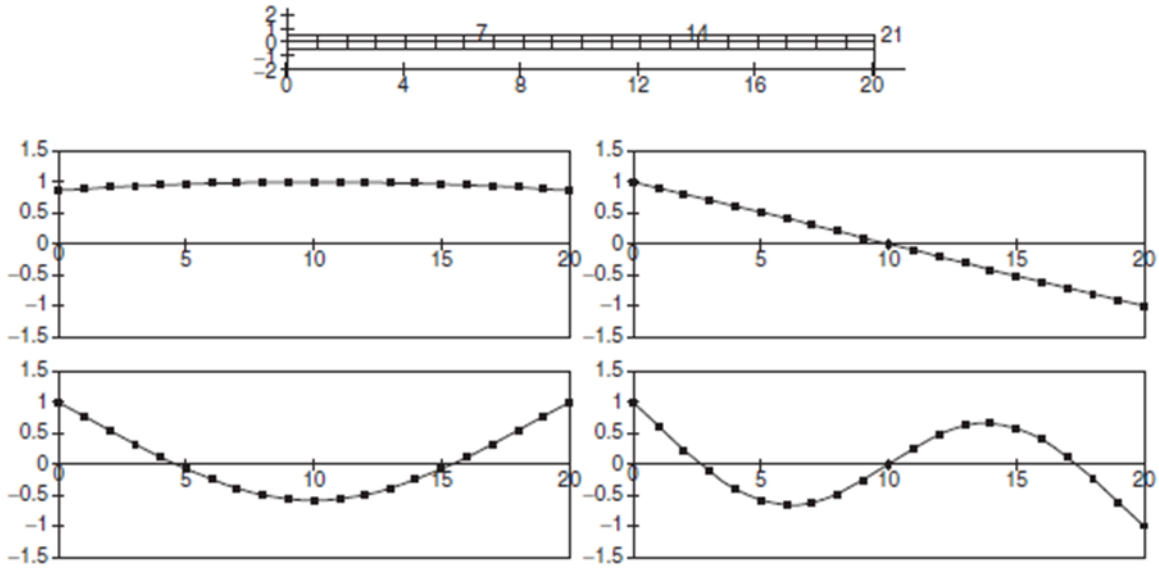

Figure 1 Model of uniform beam used for mode accuracy study, and mode shapes of the first four normal modes.

Figure 1 shows a uniform steel beam 20 in. long, 1 in. in diameter, and with an undamped bearing at each the end having K = 1000 lb/in. The mode shapes of the first four modes are also shown. The first two modes are fairly rigid. Modes like these don't require many elements for good accuracy. The second two modes look like a half sine wave and a full sine wave.

Figure 2 shows that accuracy depends not only on the number of elements, but also on the formulation of the elements. Notice in Figure 2 that the rigid modes have very small percentage errors. This is because there is little to no flexure of the elements. The flexible modes, however, illustrate the effect of the number of elements and also the difference between the four different element types. For our uniform beam test rotor, the four-element case corresponds to an L/D of 5. For this case the first mode, which is a rigid shaft mode, has frequency errors less than 1 percent. Even the second mode, which is also a rigid shaft mode, has errors less than 1 percent for all but the uncorrected 2-node element. For the first flexible mode, however, four elements are not enough, and the errors exceed 10 percent. Based on the errors computed for this example, a guideline of "use elements with L/D < 1" will result in acceptable accuracy for all cases considered. Another reason to maintain an element L/D < 1 is that when using a transfer matrix-based

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calculation procedure, the calculations can become numerically ill-conditioned if the L/D of very many elements is much longer than one. This is an aspect specific to the algorithm, and not the rotor system model.

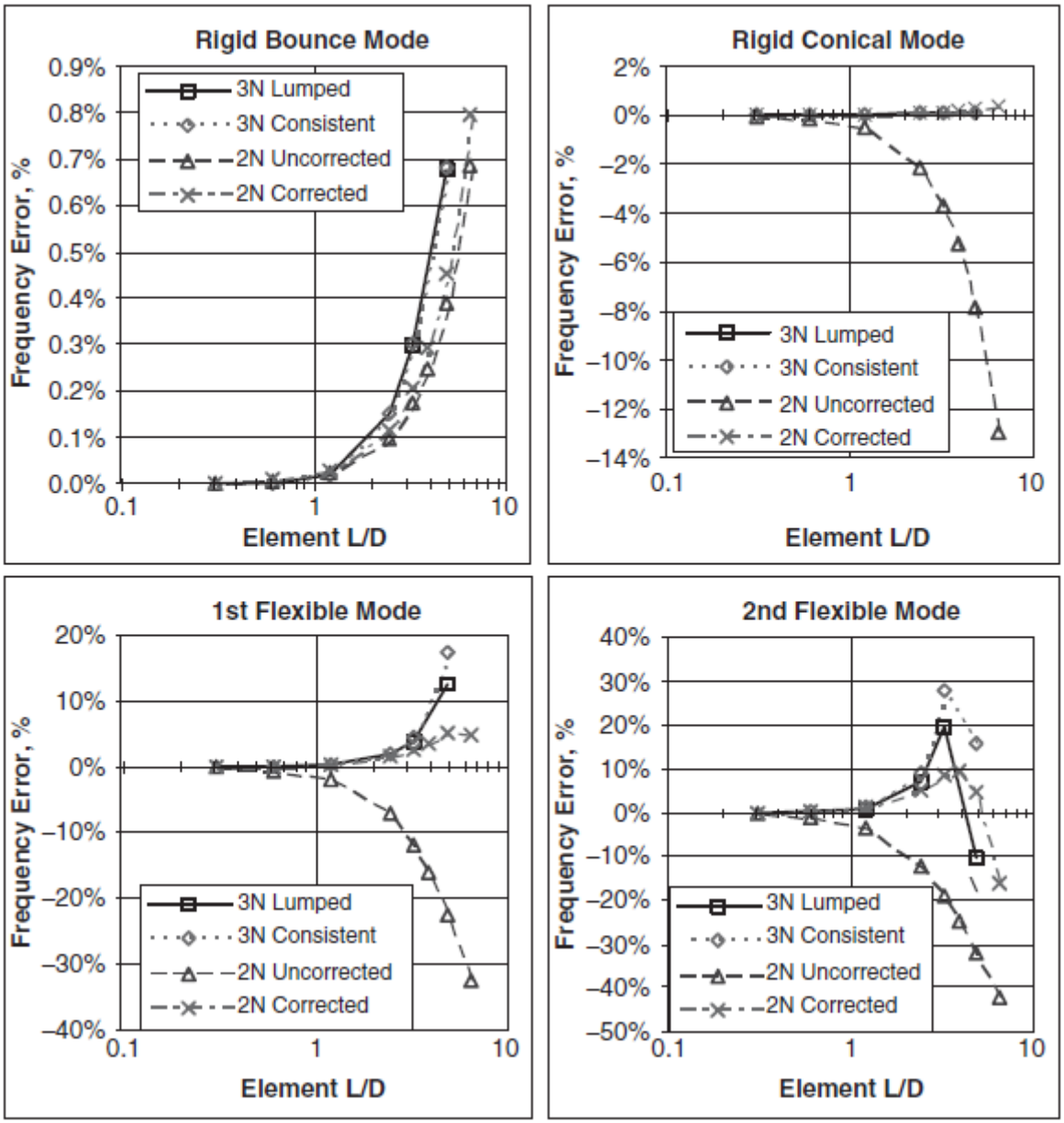

Figure 2 Mode accuracy versus number of elements for four different element definitions.

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7.2 45 degrees rule

A common situation encountered in rotor modeling is a large abrupt change in diameter. The bending stiffness of the shaft is not able to change quite as sharply as the diameter. A step change in diameter of one-third or more is considered abrupt. The material outside the 45° line shown in Figure 3 will not contribute significantly to the bending stiffness of the section. Two ways to model this section are shown, one using cylindrical beams and one using conical beams. Both approaches model the transition section with a layered pair of beams (inner and outer layers). The inner beam has the actual density and modulus of the shaft material. The outer beam has the actual density, but zero modulus so as to not contribute any stiffness.

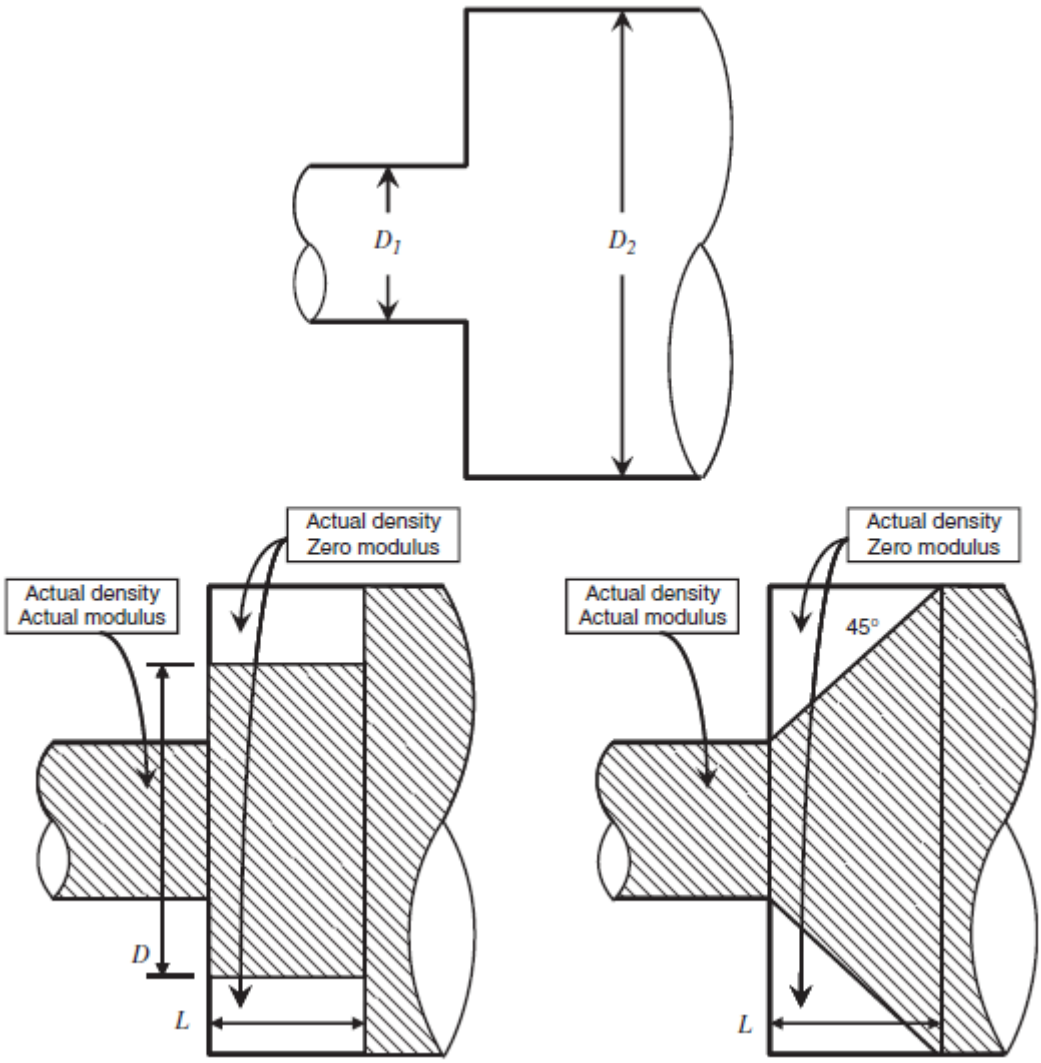

Figure 3 Two methods of the 45-degree rule for modeling abrupt diameter changes.

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7.3 Stations for Added Weight.

The inertia properties of pump impellers are most often added to the rotor shaft model as added concentrated weights and inertias. When constructing the model of the underlying shaft, be sure to provide stations at or close to the center of gravity of each impeller. These will be the locations where the concentrated masses will be placed.

7.4 Fluid "Added Mass".

The fluid surrounding the rotor adds inertia to the rotor in two ways:

- the fluid trapped in the impeller passages adds mass directly, and this can be calculated based on the volume in the impeller passages times the pumped fluid density;
- the fluid around the periphery of the impellers that is displaced by the vibrating motion of the impeller; this part of the added mass is equal to the "swept volume" of the impellers and immersed shafting, times the density of the pumped liquid.

References

Turbomachinery Rotordynamics – Phenomena, Modelling & Analysis – Dara Childs

Machinery Vibration & Rotordynamics – John Vance - Fouad Zeidan - Brian Murphy

API STANDARD 610, 12th Edition, January 2021

API PUBBLICATION 684, 1st Edition, February 1996

ANSI/HI 9.6.8-2014 Rotordynamic Pumps – Guideline for Dynamic Pumping Machinery

G.10.00.09

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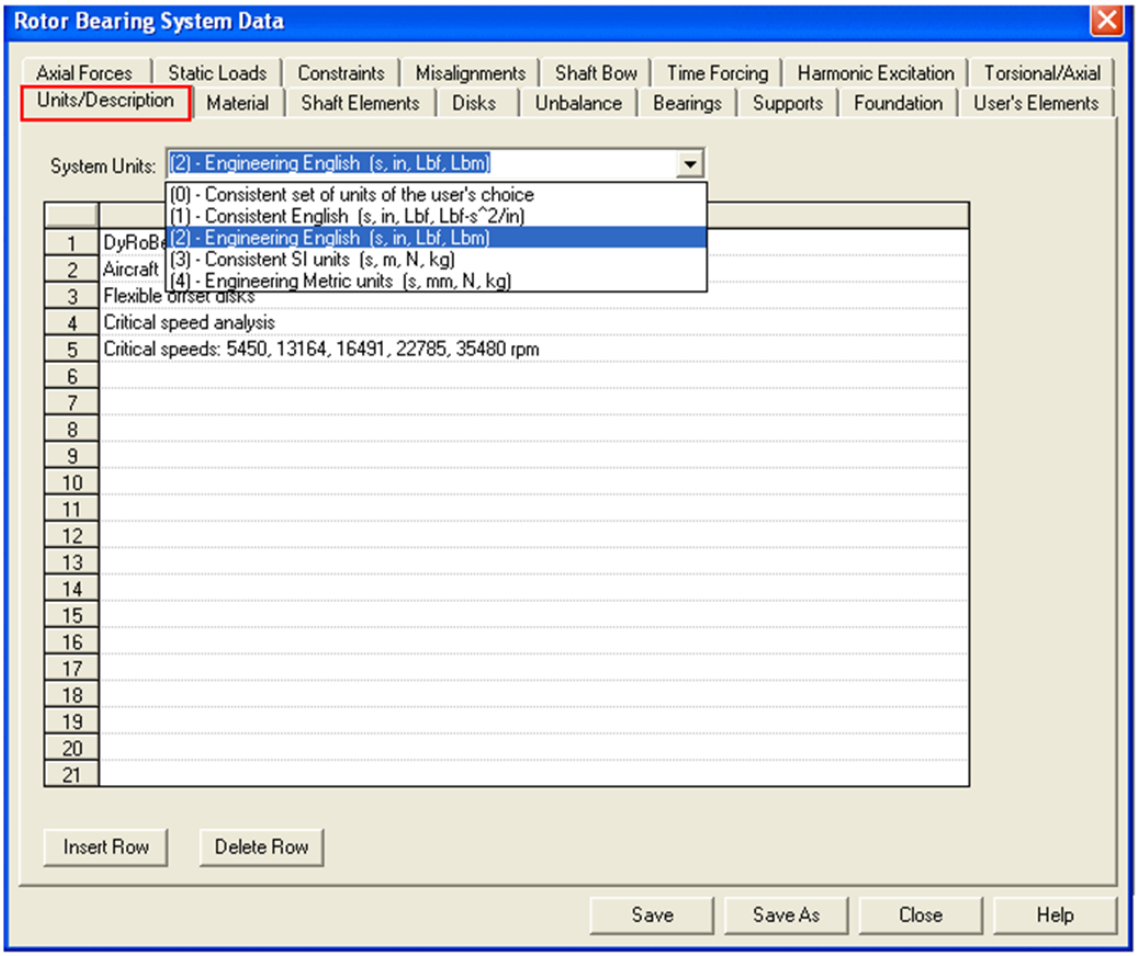
Annex

Annex 1 Uses of DyRoBes.



Step1: unit system.

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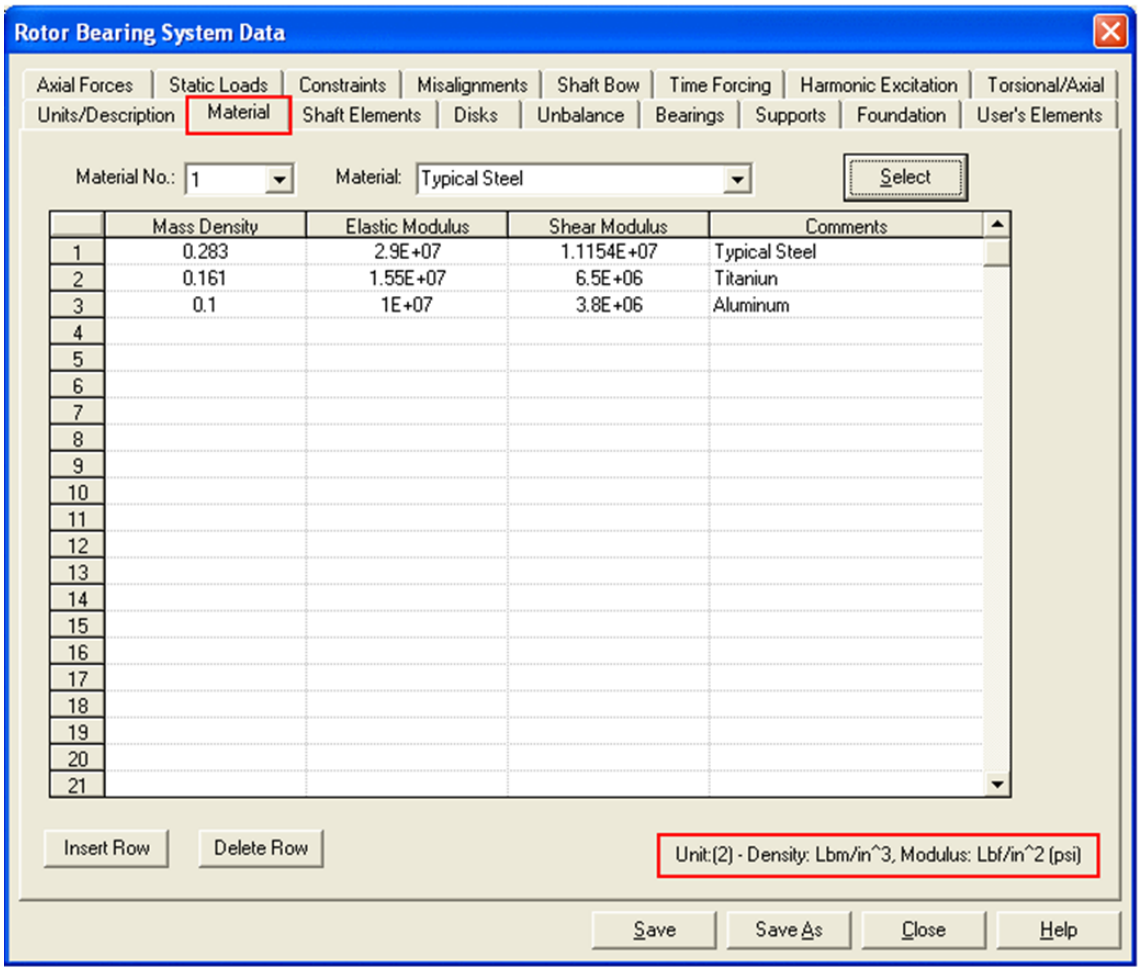

Step 2: material.

Enter the density, elastic and shear modulus of the materials of the shaft. Include a “dummy” material with very low elastic and shear modulus to model the rotors parts with no stiffness contribution.

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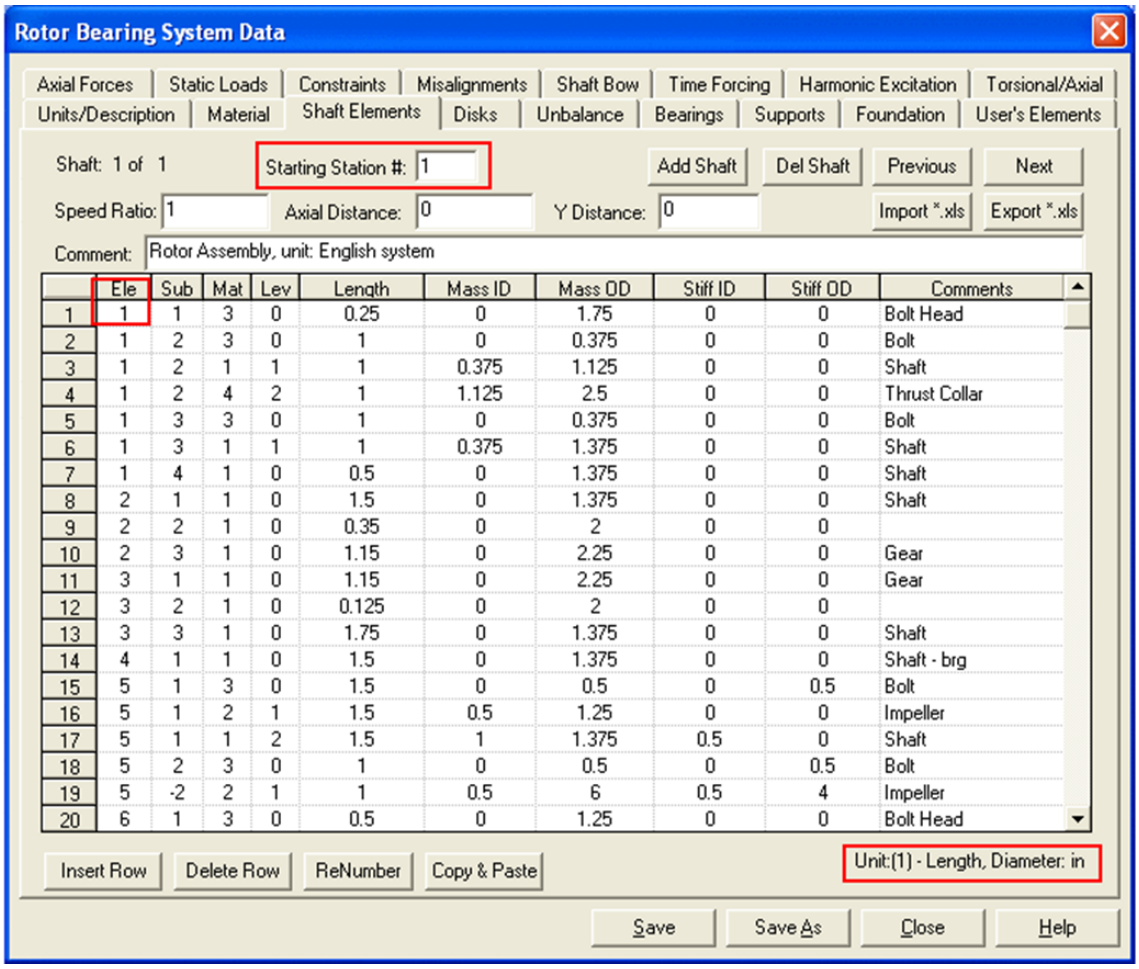

	Ele	Sub	Mat	Lev	Length	Mass ID	Mass OD	Stiff ID	Stiff OD	Comments
1	1	1	3	0	0.25	0	1.75	0	0	Bolt Head
2	1	2	3	0	1	0	0.375	0	0	Bolt
3	1	2	1	1	1	0.375	1.125	0	0	Shaft
4	1	2	4	2	1	1.125	2.5	0	0	Thrust Collar
5	1	3	3	0	1	0	0.375	0	0	Bolt
6	1	3	1	1	1	0.375	1.375	0	0	Shaft
7	1	4	1	0	0.5	0	1.375	0	0	Shaft
8	2	1	1	0	1.5	0	1.375	0	0	Shaft
9	2	2	1	0	0.35	0	2	0	0	
10	2	3	1	0	1.15	0	2.25	0	0	Gear
11	3	1	1	0	1.15	0	2.25	0	0	Gear
12	3	2	1	0	0.125	0	2	0	0	
13	3	3	1	0	1.75	0	1.375	0	0	Shaft
14	4	1	1	0	1.5	0	1.375	0	0	Shaft - brg
15	5	1	3	0	1.5	0	0.5	0	0.5	Bolt
16	5	1	2	1	1.5	0.5	1.25	0	0	Impeller
17	5	1	1	2	1.5	1	1.375	0.5	0	Shaft
18	5	2	3	0	1	0	0.5	0	0.5	Bolt
19	5	-2	2	1	1	0.5	6	0.5	4	Impeller
20	6	1	3	0	0.5	0	1.25	0	0	Bolt Head

Step 3: shaft elements

The shafts of the rotor system are numbered consecutively from 1 to Ns and the finite element stations of the model are numbered consecutively starting with 1 at the left end of shaft 1 and continuing to the last station at the right end of shaft Ns. The shafts are made up of Elements with the numbering for each shaft starts at the left end. Stations are located at the ends of the Elements. The term Station is commonly used in the rotordynamics instead of the term Node which is generally used in finite element literature because of the alternate meaning that Node has in the vibration mode shapes. Two types of elements are included in DyRoBeS:

- cylindrical Element;
- conical (Tapered) Element.

Element *i* is located immediately to the right of station i. Each element may possess several subelements (starting with 1 at the left of the element) thereby allowing for reasonable flexibility in modeling systems with several geometric discontinuities. Each subelement may possess several levels or layers (starting with zero). When using subelements, it should be kept in mind that Disks and Bearings can only be placed at the ends of the elements (stations).

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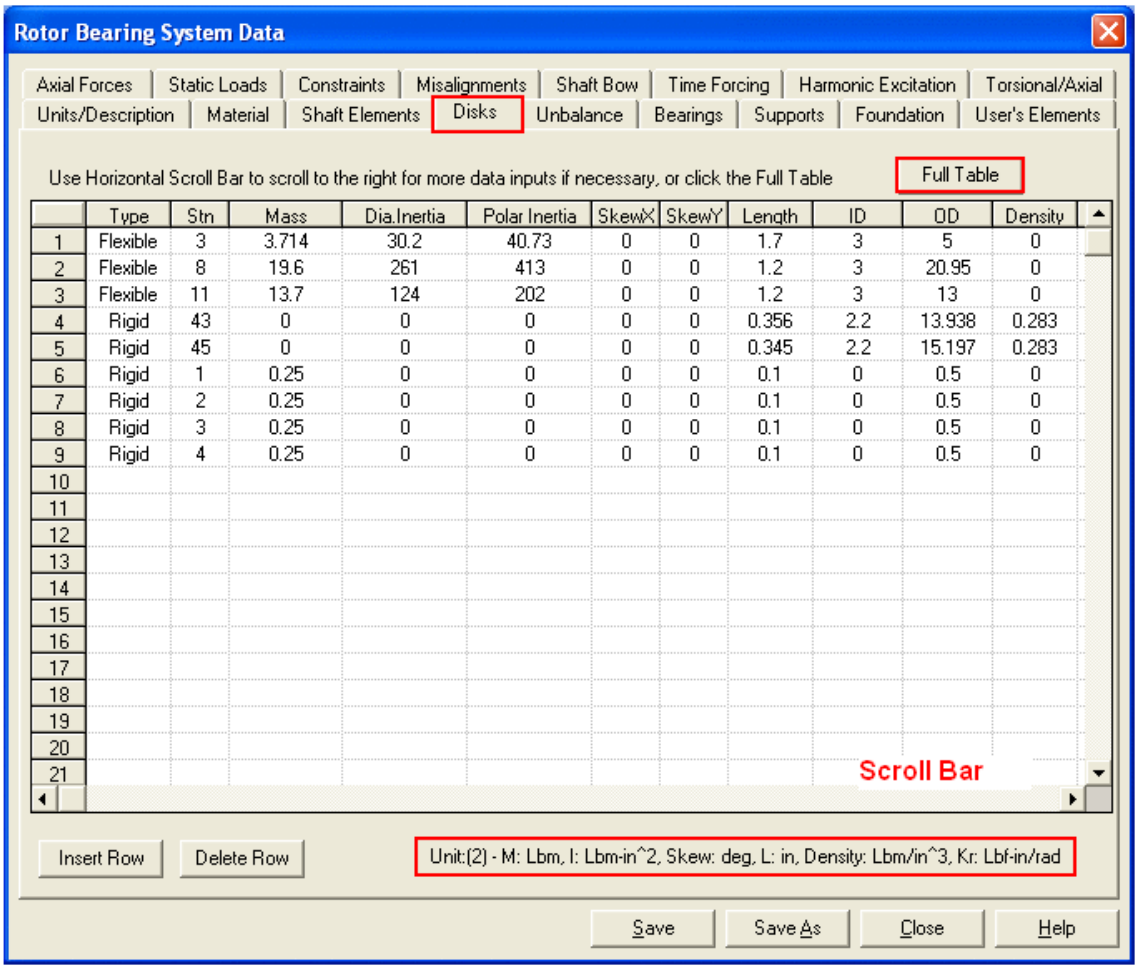

	Type	Stn	Mass	Dia.Inertia	Polar Inertia	SkewX	SkewY	Length	ID	OD	Density
1	Flexible	3	3.714	30.2	40.73	0	0	1.7	3	5	0
2	Flexible	8	19.6	261	413	0	0	1.2	3	20.95	0
3	Flexible	11	13.7	124	202	0	0	1.2	3	13	0
4	Rigid	43	0	0	0	0	0	0.356	2.2	13.938	0.283
5	Rigid	45	0	0	0	0	0	0.345	2.2	15.197	0.283
6	Rigid	1	0.25	0	0	0	0	0.1	0	0.5	0
7	Rigid	2	0.25	0	0	0	0	0.1	0	0.5	0
8	Rigid	3	0.25	0	0	0	0	0.1	0	0.5	0
9	Rigid	4	0.25	0	0	0	0	0.1	0	0.5	0
10											
11											
12											
13											
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15											
16											
17											
18											
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21											

Step 4: disc.
A concentrated disk can be located at any finite element station. Disks cannot be placed at a substation, and must be placed at stations. Multiple disks are allowed at the same station.

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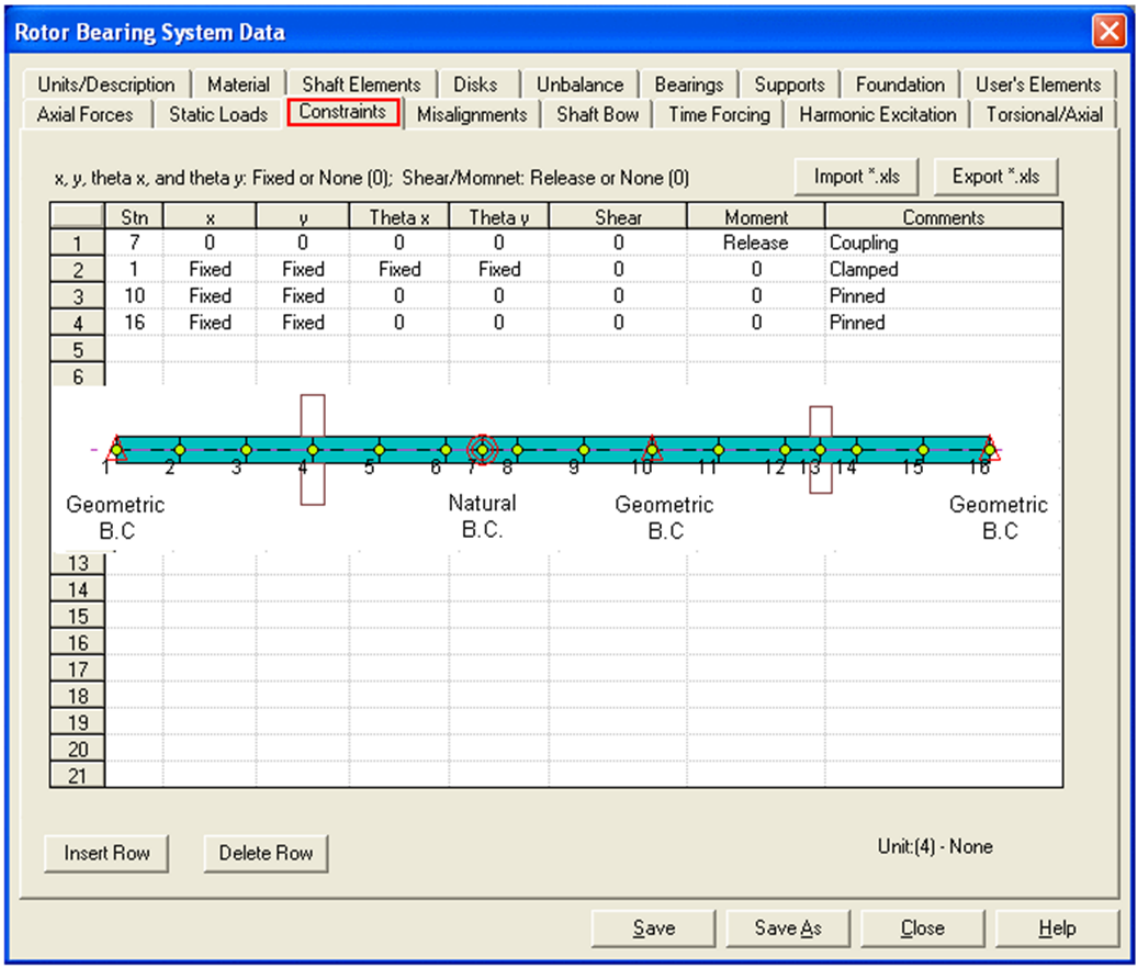

Step 5: constraints.
Add constraint (fixed X and Y) to all bearing stations.

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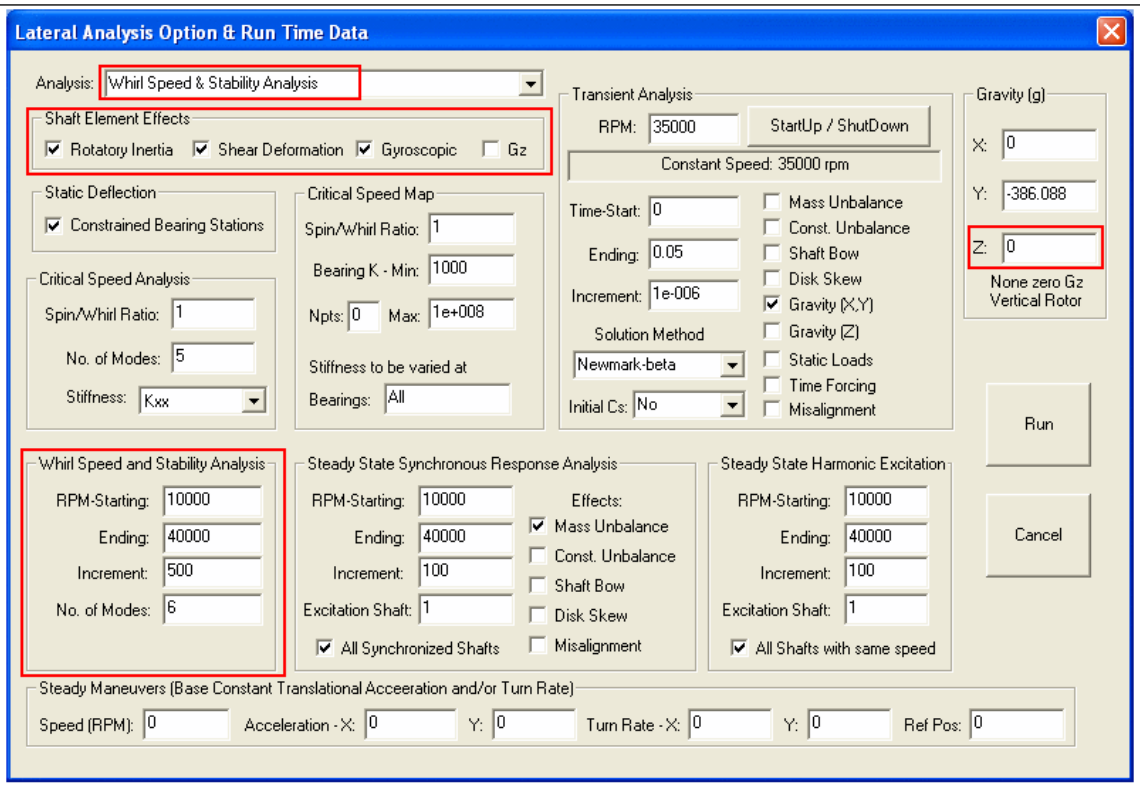
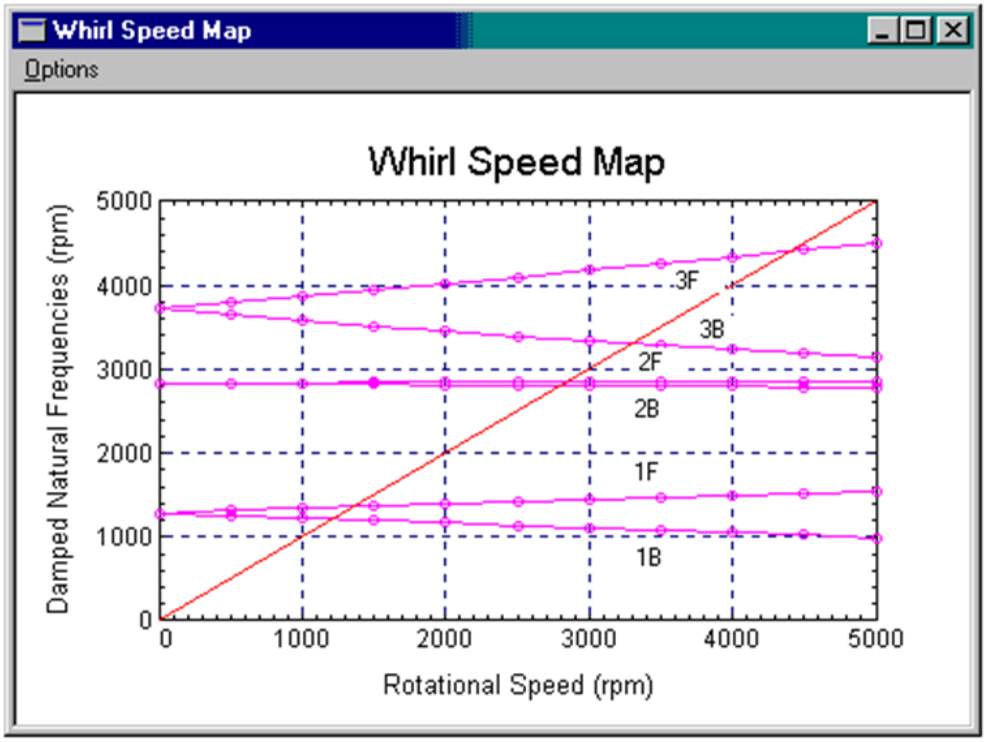


Step 6: analysis.

Complete a Whirl Speed & Stability Analysis:

- RPM-Starting: 0% of maximum speed;
- ending: greater than 120% of maximum speed (130% if the pump can run dry);
- number of Modes: at least two



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Step 7: postprocessing.
Create a Whirl Speed Map.
The excitation slope must be 1.0 (red line); it is defined under “Options” in the graph.
If all the intersection between red line (synchronous excitations) and pink lines (natural frequencies) has an abscissa greater than 120% of maximum speed (130% in case rotor is designed to run dry), the pump is verified

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